

Devi Madhuri Donthra Boina

Senior Data Engineer AI

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SUMMARY

Senior Data Engineer with around 5+ years of experience in building and operating production grade ETL/ELT pipelines on AWS and Spark ecosystems. Delivered event driven pipelines that cut runtime by 30% and added end to end observability that reduced incident response time by 35%. Now focused on moving research models to production and tightening IAM policies for least privilege security.

SKILLS

Cloud & MLOps: AWS, Azure, EC2, S3, EMR, Lambda, Glue, Redshift, Athena, Kinesis, SQS, SNS, Step Functions, CloudWatch, Cloud storage

Data Engineering & Big Data: PySpark, Apache Spark, Spark Structured Streaming, Kafka, Apache Iceberg, Airflow, dbt, ETL, ELT, CDC, Event Driven Architecture, Stream Processing, Batch Processing, Hadoop, SSIS, Data warehousing

Databases & Querying: PostgreSQL, DynamoDB, Snowflake, SQL, Scala, NoSQL, Netezza

Machine Learning & GenAI: LangChain, RAG, LLM

Programming & Tools: Python, Pandas, NumPy, Bash, Flask, Django, Spring Boot, Terraform, Docker, Kubernetes, CI/CD, IAM, RBAC

Business Intelligence & Visualization: Splunk

Data Governance & Quality: Data Quality, Data Lineage, Data Observability, Data Modeling, Data Governance

Project Management & Collaboration: JIRA, Agile delivery

EXPERIENCE

STATE FARM

Chicago, IL

AWS DATA ENGINEER

Jul 2024 – Present

- Designed and deployed scalable ETL pipelines using AWS Glue, PySpark, and Python, integrating with SQL, NoSQL, and Cloud storage, while ensuring near complete coverage of data warehousing and ELT processes, resulting in a 25% reduction in data processing time.
- Designed and optimized production ETL pipelines using AWS Glue, EMR, PySpark, Scala, and Python, ingesting data from S3, Apache Iceberg, and Glue catalog tables, reducing pipeline runtime by 30% through improved partitioning and parallel execution.
- Utilized Terraform to manage cloud infrastructure, provisioning resources such as Azure, Hadoop, and Netezza, while implementing SSIS for data integration and ensuring compliance with organizational security policies, resulting in a 30% reduction in infrastructure costs.
- Implemented end to end observability with Splunk, reducing incident response time by 35% and ensuring reliable data delivery across business critical pipelines in a regulated insurance environment.
- Enforced least privilege IAM security policies across Lambda, Glue, API Gateway, and cross account S3, configuring STS role assumption for CI/CD pipelines to eliminate long lived credentials and comply with requirements.
- Improved analytical query performance by 20% by using Redshift Spectrum and Starburst as federated query engines for Apache Iceberg tables in S3, enabling cross source SQL analytics without additional ETL overhead.
- Implemented data quality monitoring and anomaly detection with AWS CloudWatch and custom Python scripts, reducing pipeline errors by 25% and maintaining data accuracy across 10+ daily scheduled runs.
- Collaborated with solution architects, product engineers, and analytics leads to translate complex data integration requirements into scalable AWS pipeline solutions that align with business and compliance needs.
- Collaborated with stakeholders to implement Agile delivery methodologies, utilizing JIRA to track progress and ensure timely completion of projects, while mentoring team members on best practices for collaboration, knowledge sharing, and stakeholder management.
- Developed and implemented data quality monitoring and anomaly detection using AWS CloudWatch, Python, and SQL, resulting in a 20% reduction in data errors and ensuring reliable data delivery across business critical pipelines.

BANK OF AMERICA

Hartford, CT

DATA ENGINEER

Aug 2022 – Jun 2024

- Architected and led the migration of a large scale data platform to Delta Lake on Databricks, utilizing AWS Glue, Python, and Scala to ensure seamless data integration, while implementing ETL and ELT processes using Spark, resulting in a 40% improvement in data processing efficiency.
- Designed and optimized Snowflake data pipelines using Streams, Tasks, and dynamic data masking, improving query performance by 25% and enforcing role based access control for compliance sensitive financial datasets.
- Built and maintained production ETL pipelines on AWS using Glue, Lambda, and Step Functions to ingest, transform, and load financial data from multiple source systems, supporting 15+ daily scheduled runs.
- Optimized Spark and SQL jobs with partition pruning, broadcast joins, predicate pushdown, and caching, reducing average job runtime by 35%. Improved MS SQL Server stored procedures and indexes to enhance ETL and reporting pipeline performance.
- Developed and deployed RESTful microservices using Flask and Spring Boot, containerized with Docker and orchestrated on Kubernetes, enabling independent scaling and zero downtime deployments for consuming teams.
- Redesigned IAM policies for AWS Glue, Lambda, DynamoDB, and cross account S3, implementing STS role assumption in Jenkins CI/CD pipelines to eliminate static credentials and reduce model deployment time by 40%.
- Authored DBT models to centralize transformation and business logic for finance and compliance reporting, and implemented automated Python validation and AWS CloudWatch monitoring to maintain data accuracy across all pipeline runs.
- Designed and implemented a scalable data warehousing solution using Snowflake, SQL, and NoSQL, while ensuring data governance and compliance with organizational security policies, resulting in a 25% reduction in data storage costs.

DENTSU

Hyderabad, India

PYTHON DEVELOPER

Aug 2020 – Dec 2021

- Developed and maintained scalable ETL pipelines using Python, PySpark, and Azure Data Factory, integrating with SQL, NoSQL, and Cloud storage, while ensuring near complete coverage of data integration and data quality processes, resulting in a 30% improvement in data processing efficiency.
- Designed and managed multi step workflow orchestration using Apache Airflow with Python based DAGs, reducing manual intervention by automating dependency management and maintaining pipeline consistency across environments.
- Developed custom Python ingestion scripts and SQL queries to process multiple data formats, including CSV, JSON, XML, Excel, and Parquet, and implemented source specific parsing, validation, and transformation logic for each system.
- Built REST APIs with Flask and Django to enable self service data access, allowing business teams to query and retrieve data independently and significantly reducing ad hoc engineering requests.
- Implemented large scale batch processing with Python and the Spring Batch framework, automated deployments via CI/CD pipelines, and optimized query execution logic, improving overall processing speed by 30%.
- Configured monitoring and alerting dashboards with Python scripts and Azure Monitor, improving pipeline observability and shifting the team from reactive to proactive incident response for data anomalies and failures.
- Collaborated with cross functional teams to design and implement a data governance framework, utilizing Azure, Hadoop, and Netezza to ensure data quality, security, and compliance, resulting in a 20% reduction in data related risks.

EDUCATION

NEW ENGLAND COLLEGE

Henniker, NH

Master of Science in Computer and Information Systems

Jan 2022 – May 2023

- GPA: 3.95

CVR COLLEGE OF ENGINEERING

India

Bachelor of Technology in Electronics and Communication Engineering

Aug 2016 – Apr 2020

- GPA: 3.56

CERTIFICATIONS

- AWS Certified Cloud Practitioner: Issued 2023
- AWS Certified Developer, Associate: Issued 2025